Rockchip RV1106 Datasheet

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Revision History

Date	Revision	Description
2026-06-02	2.2	Update package thermal characteristics of RV1106 in Table 4-1
2026-04-02	2.1	• Update package information of RV1106G2, RV1106G3 and RV1106 in 2.1 • Correct absolute and recommended supply power of DDR IO in Table 3-1 and Table 3-2
2026-04-02	2.0	Add RV1106 spec in this document, detail as below: • In 1.2, add features of video codec, ISP, SARADC and package type of RV1106. Correct NPU performance as 1 TOPS. Update rate of external off-chip memory. • In 1.3, correct external memory interface to fit RV1106 • In 2.1, add order information of RV1106 • In 2.2, add marking of RV1106 • In 2.3, add POD information of RV1106 • In 2.6, add pin number list of RV1106 Add internal on-chip memory feature of RV1106G3 in 1.2.2
2025-12-12	1.9	Modify the names from pin 117 to 128
2025-03-12	1.8	Modify the names of pin 105 and 110
2023-12-18	1.7	Update the operating temperature range
2023-08-28	1.6	Update the RV1106G3 feature description
2023-07-25	1.5	Update the operating temperature range
2023-07-18	1.4	Update the description about DVP interface
2023-05-22	1.3	Update Table 3-1 and 3-2 information
2022-12-12	1.2	Add RV1106G3
2022-04-27	1.1	Update MSL information and package thermal characteristics
2022-04-06	1.0	Initial release

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Chapter 1 Introduction

1.1 Overview

RV1106 series is a highly integrated vision processor SoC for IPC, especially for AI related application.

It is based on single-core ARM Cortex-A7 32-bit core which integrates NEON and FPU. There is a 32KB I-cache and 32KB D-cache and 128KB unified L2 cache.

The build-in NPU supports INT4/INT8/INT16 hybrid operation and the computing power is up to 1 TOPS. In addition, with its strong compatibility, network models based on a series of frameworks such as TensorFlow/MXNet/PyTorch/Caffe can be easily converted.

RV1106 series introduces a new generation totally hardware-based maximum 5-Megapixel ISP (image signal processor). It implements a lot of algorithm accelerators, such as HDR, 3A, LSC, 3DNR, 2DNR, sharpening, dehaze, gamma correction and so on. Cooperating with two MIPI CSI (or LVDS) and one DVP (BT.601/BT.656/BT.1120) interface, users can build a system that receives video data from 3 camera sensors simultaneous.

The video encoder embedded in RV1106 series supports H.265/H.264 encoding. It also supports multi-stream encoding. With the help of this feature, the video from camera can be encoded with higher resolution and stored in local memory and transferred another lower resolution video to cloud storage at the same time. To accelerate video processing, an intelligent video engine with 22 calculation units is also embedded.

RV1106G2 and RV1106G3 have built-in 16-bit DRAM DDR3L, while RV1106 supports external 16-bit DDR2/DDR3/DDR3L/DDR4 memory interface. RV1106 series is capable of sustaining demanding memory bandwidths, also integrated build-in RTC, POR, audio codec and MAC PHY.

1.2 Features

1.2.1 Application Processor

- Single core ARM Cortex-A7
- Full implementation of the ARM architecture v7-A instruction set, ARM Neon Advanced SIMD
- Separately Integrated Neon and FPU
- 32KB L1 I-Cache and 32KB L1 D-Cache
- Unified 128KB L2 Cache for Cortex-A7
- TrustZone technology support
- One isolated voltage domain to support DVFS

1.2.2 Memory Organization

- Internal on-chip memory
  - BootRom
    - Support system boot from the following device:
      - SPI interface
      - eMMC interface
      - SD/MMC interface
    - Support system code download by the following interface:
      - USB interface
      - UART interface
  - 256KB Share Memory
  - 8KB PMU SRAM
  - RV1106G2 SIP 1Gb DDR3L
  - RV1106G3 SIP 2Gb DDR3L
- External off-chip memory

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- Dynamic Memory Interface
  - Compatible with JEDEC standards
  - Compatible with DDR2-1056/DDR3-2133/DDR3L-2133/DDR4-2400
  - Support 16-bit data width
  - Support 1 ranks (chip selects)
  - Support max 1024MB addressing space
  - Low power modes, such as power-down and self-refresh for SDRAM

- eMMC Interface
  - Fully compliant with JEDEC eMMC 4.51 specification
  - Support HS200, but not support CMD Queue
  - Support three data bus width: 1bit, 4bits or 8bits
- SD/MMC Interface
  - Compatible with SD3.0, MMC ver4.51
  - Data bus width is 4bits
- Flexible Serial Flash Interface (FSPI)
  - Support transfer data from/to serial flash device
  - Support 1bit, 2bits or 4bits data bus width

1.2.3 System Component

- MCU
  - MCU in VD_CORE integrate 16KB Cache
  - Integrated Programmable Interrupt Controller, all IRQ lines connected to GIC for CPU also connect to MCU
  - Integrated Debug Controller with JTAG interface

- CRU (clock & reset unit)
  - Support total 4 PLLs to generate all clocks
  - One oscillator with 24MHz clock input
  - Support clock gating control for individual components
  - Support global soft-reset control for whole chip, also individual soft-reset for each component

- PMU (power management unit)
  - Multiple configurable work modes to save power by different frequency or automatic clock gating control or power domain on/off control
  - Lots of wakeup sources in different mode
  - Support 3 separate voltage domains，VDD_ARM,VDD_LOGIC,VDD_PMU.

- Timer
  - Support 2 secure timers with 64bits counter and interrupt-based operation
  - Support 6 non-secure timers with 64bits counter and interrupt-based operation
  - Support two operation modes: free-running and user-defined count for each timer
  - Support timer work state checkable

- PWM
  - Support 12 on-chip PWMs (PWM0~PWM11) with interrupt-based operation
  - Programmable pre-scaled operation to bus clock and then further scaled
  - Embedded 32-bit timer/counter facility
  - Support capture mode
  - Support continuous mode or one-shot mode
  - Provides reference mode and output various duty-cycle waveform
  - Optimized for IR application for PWM3, PWM7, PWM11

- Watchdog
  - 32-bit watchdog counter
  - Counter counts down from a preset value to 0 to indicate the occurrence of a timeout

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- WDT can perform two types of operations when timeout occurs:
  - Generate a system reset
  - First generate an interrupt and if this is not cleared by the service routine by the time a second timeout occurs then generate a system reset
- One Watchdog for non-secure application
- One Watchdog for secure application

- Interrupt Controller
  - Support 121 SPI interrupt sources input from different components inside RV1106 series
  - Support 16 software-triggered interrupts
  - Input interrupt level is fixed, high-level sensitive or rising edge sensitive
  - Support different interrupt priority for each interrupt source, and they are always software-programmable

- DMAC
  - Micro-code programming-based DMA
  - Linked list DMA function is supported to complete scatter-gather transfer
  - Support data transfer types including memory-to-memory, memory-to-peripherals, peripherals-to-memory
  - Totally three embedded DMA controllers for peripheral system
  - Each DMAC features:
    - Support 8 channels
    - 32 hardware requests from peripherals
    - 2 interrupt output
    - Support TrustZone technology and programmable secure state for each DMA channel

- Secure System
  - Embedded one cipher engines
    - Support Link List Item (LLI) DMA transfer
    - Support SHA-1, SHA-256/224, MD5 with hardware padding
    - Support HMAC of SHA-1, SHA-256, MD5 with hardware padding
    - Support AES-128, AES-192, AES-256 encrypt and decrypt cipher
    - Support DES and TDES cipher
    - Support AES ECB/CBC/OFB/CFB/CTR/CTS/XTS/GCM/CBC-MAC/CMAC mode
    - Support DES/TDES ECB/CBC/OFB/CFB mode
    - Support up to 4096 bits PKA mathematical operations for RSA/ECC/SM2
  - Support generating random numbers, one secure only engine, another one security configurable
  - Support secure OTP
  - Support secure debug
  - Support secure OS
  - Except CPU, the other masters in the SoC can also support security and non-security mode by software-programmable
  - Some slave components in SoC can only be addressed by security master and the other slave components can be addressed by security master or non-security master by software-programmable
  - System SRAM (share memory), part of space is addressed only in security mode
  - External DDR space can be divided into 16 parts, each part can be software-programmable to be enabled by each master

- Mailbox
  - One Mailbox in SoC to service CPU and MCU communication
  - Support four mailbox elements, each element includes one data word, one command word register and one flag bit that can represent one interrupt
  - Provide 32 lock registers for software to use to indicate whether mailbox is occupied

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- Decompression
  - Support for decompressing GZIP files
  - Support for decompressing data in DEFLATE format
  - Support for decompressing data in ZLIB format
  - Support the limit size function of the decompressed data to prevent the memory from being maliciously destroyed during the decompression process

- Real Time Clock (RTC)
  - Provides year, month, day, weekday, hours, minutes, and seconds based on a 32.768 kHz crystal oscillator
  - Support compensation for the second and hour count
  - BCD representation of time, calendar and alarm
  - 12- or 24-hour clock with AM and PM in 12-hour mode
  - Interrupts are separately software maskable
    - Alarm interrupt
    - Periodic interrupt
    - Chip power off interrupt
    - Battery power atypical interrupt

1.2.4 Video CODEC

- Video Encoder
  - H.265/HEVC Main Profile, level 5.0
  - H.264/AVC High Profile, level 5.0
  - Support multi-channel encoding with the following performance:
    RV1106G2: up to 5-megapixel@30FPS
    RV1106G3: up to 8-megapixel@15FPS
    RV1106: up to 8-megapixel@15FPS
  - JPEG baseline, up to 4-megapixel @60fps in standalone mode, resolution up to 8192 x 8192
  - Bitrate up to 60Mbps
  - Six bit rate control modes (CBR, VBR, FIXQP, AVBR, QPMAP, and CVBR)
  - Support YUV420 and YUV400 format input
  - Intelligent encoding mode
  - 8-area OSD
  - YUV/RGB video source with crop, rotation and mirror
  - Ultra-low delay encoding

1.2.5 Neural Process Unit

- Neural network acceleration engine is 1 TOPS
- Support integer 4, integer 8 and integer 16 operation
- Support creating simple custom operators
- Support deep learning frameworks: TensorFlow, Caffe, Tflite, Pytorch, Onnx NN, etc.

1.2.6 Rockchip Intelligent Video Engine (RKIVE)

- GMM
  - Support 1 to 5 gaussian model
- BGM (base on codebook)
  - Support 3 codebook model
- Canny
  - Staging buffer stride require 64 pixel align
  - Support 3X3 and 5x5 template coefficient
- CCL
  - Support max to 254 connected regions
  - Support 4-connected and 8-connected region
- Stcorner
  - Support max to 500 corner sort output
- LK

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  - Support max to 500 corner input
  - Support 1~4 optical flow layers
- Integral
  - Require all the buffer base is 16bytes align
- LBP
  - Support simple and absolute value comparison mode
- Filter
  - Support 3X3 and 5x5 mode
- Sobel
  - Support 3X3 and 5x5 mode
- Morph
  - Support eroding and dilating mode
- Denoise Filter
  - Support minimum/median/maximum 3 types filter
- DMA
  - Support direct copy mode
  - Support interval copy mode
- CSC
  - Support rgb2yuv, yuv2rgb, rgb2hsv, yuv2hsv
  - Support 601 and 709 format, full and limit range
- Hist/eqhist
  - Support hist only, eqhist only, hist + eqhist 3 types mode
- Logic OP
  - Support logic and, logic or, logic xor, add, sub, absolute difference
- Mag and Ang
  - Calculation of the image gradient magnitude and direction
- Morph
  - Support eroding and dilating mode
- NCC
  - Calculation of the image normalized cross-correlation
- Cast
  - Data linear transformation
- Sad
  - Support sad size is 4x4, 8x8 and 16x16
- Threshold
  - Convert grayscale into a binary image
- Map
  - Support 8bit to 8bit and 8bit to 16bit map operation

1.2.7 Graphics Engine

- 2D Graphics Engine
  - Input data:
    - ARGB/RGB888/RGB565/RGB4444/RGB5551
    - YUV420/YUV422/YVYU422/YVYU420/YUV422SP10bit/YUV420SP10bit
  - Output data:
    - ARGB/RGB888/RGB565/RGB4444/RGB5551
    - YUV420/YUV422/YUV400/Y4/YVYU422/YVYU420
  - Pixel Format conversion, BT.601/BT.709
  - Dither operation
  - Max resolution: 8192x8192 source, 4096x4096 destination
  - Scaling
    - Down-scaling: Average filter
    - Up-scaling: Bi-cubic filter(source>2048 would use Bi-linear)
    - Arbitrary non-integer scaling ratio, from 1/16 to 16
  - Rotation
    - 0, 90, 180, 270-degree rotation
    - x-mirror, y-mirror
    - Mirroring and rotation co-operation

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- BitBLT
  - Block transfer
  - Color palette/Color fill, support with alpha
  - Transparency mode (color keying/stencil test, specified value/value range)
  - Two source BitBLT
  - A+B=B only BitBLT, A support rotate & scale when B fixed
  - A+B=C second source (B) has same attribute with (C) plus rotation function
- Alpha Blending
  - Comprehensive per-pixel alpha(color/alpha channel separately)
  - Fading
  - Support SRC1(R2Y)+SRC0(YUV) -> DST(YUV)
  - Support DST Full CSC convert for YUV2YUV
- OSD Automatic Inversion
  - Supports OSD sources in ARGB8888/ARGB1555/ARGB444/ARGB2BPP format
  - Support SRC0 and OSD overlay
- Support square mosaic patterns to cover rectangular mosaic areas

1.2.8 Video Input Interface

- MIPI Interface
  - Two MIPI CSI DPHY
    - Each MIPI DPHY V1.2, 2lanes, 1.5Gbps per lane
    - Support to combine 2 DPHY together to one 4lanes
- DVP interface
  - One 8/10/12/16-bit standard DVP interface, up to 150MHz input data
  - Support BT.601/BT.656 and BT.1120 VI interface
  - Support the polarity of pixel_clk, hsync, vsync configurable

1.2.9 Image Signal Processor

- Video Capture (VICAP)
  - Support BT601 YCbCr 422 8bit input, RAW 8/10/12bit input
  - Support BT656 YCbCr 422 8bit progressive/interlace input
  - Support BT1120 YCbCr 422 16bit progressive/interlace input, single/dual-edge sampling
  - Support YUYV sequence configurable
  - Support the polarity of hsync and vsync configurable
  - Support receiving two interfaces of MIPI CSI /LVDS, up to four IDs for each interface
  - Support five CSI data formats: RAW8/10/12/14, YUV422
  - Support three modes of MIPI CSI HDR: virtual channel mode, identification code mode, line counter mode
  - Support four LVDS data formats: RAW8/10/12, YUV422
  - Support reducing frame rate
  - Support window cropping
  - Support RAW data through to ISP
  - Support 8/16/32 times down-sampling for RAW data
  - Support virtual stride when write to DDR
  - Support NV16/NV12/YUV400/YUYV output format for YUV data
  - Support compact/non-compact output format for RAW data
- Maximum input:
  RV1106G2: 3072x1728 (5M) @30FPS
  RV1106G3: 3840x2160 (8M) @15FPS
  RV1106: 3840x2160 (8M) @15FPS
- Minimum input: 256x256
- 3A: Include Auto Enhance (AE)/Histogram, Auto Focus (AF), and Auto White Balance (AWB) statistics output
- EXPANDER: Sensor expander
- BLC: Black Level Correction
- DPCC: Static/Dynamic Defect Pixel Cluster Correction

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- PDAF: Phase Detection Auto Focus
- LSC: Lens Shading Correction
- Bayer-2DNR: Spatial Bayer-raw Noise Reduction
- Bayer-3DNR: Temporal Bayer-raw Noise Reduction
- CAC: Chromatic Aberration Correction
- HDR-MGE: 2-Frame Merge into High-Dynamic Range
- HDR-DRC: HDR Dynamic Range Compression, Tone mapping
- GIC: Green Imbalance Correction
- DeBayer: Advanced Adaptive Demosaic
- CCM/CSM: Color Correction Matrix, RGB2YUV, etc.
- Gamma: Gamma out correction
- Dehaze/Enhance: Automatic Dehaze and effect enhancement
- 3DLUT: 3D-LUT Color Palette for Customer
- LDCH: Lens Distortion Correction only in the Horizontal direction
- YUV-2DNR: Spatial YUV Noise Reduction
- Sharp: Image sharpening and boundary filtering
- CMSK: Privacy cover and mask
- Gain: Image local gain
- Multi-sensor reuse ISP, 4 sensors for maximum
- Bus interface: 32bit AHB configuration, 128bit AXI R/W
- Low power, auto-gating for each block
- MI R/W burst group to improve memory utilization
- MI 3+2 path output, MP stepless scaling, SP/BP scaling under 1080p, MPDS/SPDS fixed 1/16 downscaling

1.2.10 Display interface

- Parallel RGB LCD Interface: 18-bit (RGB666), 16-bit (RGB565)
- Serial RGB LCD Interface
- MCU LCD Interface
- Max output resolution: 1280x720 for RGB/BT656/BT1120

1.2.11 Video Output Processor

- Display process
  - Background layer
    - programmable 18-bit color
  - Win1 layer
    - RGB888, ARGB888, RGB565
    - Support virtual display
    - 256 level alpha blending (pre-multiplied alpha support)
    - Transparency color key
    - RGB2YCbCr (BT601/BT709)
- Others
  - Support RGB or YUV domain overlay
  - BCSH (Brightness, Contrast, Saturation, Hue adjustment)
  - BCSH: YCbCr2RGB (rec601-mpeg/ rec601-jpeg/rec709)
  - BCSH: RGB2YCbCr (BT601/BT709)
  - Support dither down allegro RGB888to666 RGB888to565 & dither down frc (configurable) RGB888to666
  - Blank and black display
  - Standby mode
  - Support all layers reg done separately

1.2.12 Audio Interface

- I2S0 with 8 channels
  - Up to 8 channels TX and 8 channels RX path
  - Audio resolution from 16bits to 32bits
  - Sample rate up to 192KHz
  - Provides master and slave work mode, software configurable

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- Support 3 I2S formats (normal, left-justified, right-justified)
- Support 4 PCM formats (early, late1, late2, late3)

- Audio Codec
  - Support two 24-bits ADC channels with 90dB SNR for stereo recording from microphone
  - Support one 24-bits DAC channels with 90dB SNR for stereo playback
  - Support differential and single-ended microphone or line input
  - Sampling rate of 8KHz/12KHz/16KHz/24KHz/32KHz/44.1kHz/48KHz/96KHz

1.2.13 Connectivity

- SDIO interface
  - Compatible with SDIO3.0 protocol
  - 4-bit data bus widths

- MAC 10/100M Ethernet controller and embedded PHY
  - Support one Ethernet controllers
  - Support 10/100-Mbps data transfer rates with the RMII interfaces
  - Support both full-duplex and half-duplex operation

- USB 2.0
  - Compatible with USB 2.0 specification
  - Support one USB 2.0 Host/Device
  - Supports high-speed(480Mbps), full-speed(12Mbps) and low-speed(1.5Mbps) mode
  - Support Enhanced Host Controller Interface Specification (EHCI), Revision 1.0
  - Support Open Host Controller Interface Specification (OHCI), Revision 1.0a

- SPI interface
  - Support 2 SPI Controllers (SPI0-SPI1)
  - Support two chip-select output
  - Support serial-master and serial-slave mode, software-configurable

- I2C Master controller
  - Support 5 I2C Master(I2C0-I2C4)
  - Support 7bits and 10bits address mode
  - Software programmable clock frequency
  - Data on the I2C-bus can be transferred at rates of up to 100k bits/s in the Standard-mode, up to 400k bits/s in the Fast-mode

- UART interface
  - Support 6 UART interfaces (UART0-UART5)
  - Embedded two 64-byte FIFO for TX and RX operation respectively
  - Support 5bit, 6bit, 7bit, 8bit serial data transmit or receive
  - Standard asynchronous communication bits such as start, stop and parity
  - Support different input clock for UART operation to get up to 4Mbps baud rate
  - Support auto flow control mode for all UART

1.2.14 Others

- Multiple groups of GPIO
  - All of GPIOs can be used to generate interrupt
  - Support level trigger and edge trigger interrupt
  - Support configurable polarity of level trigger interrupt
  - Support configurable rising edge, falling edge and both edge trigger interrupt
  - Support configurable pull direction (a weak pull-up and a weak pull-down)
  - Support configurable drive strength

- Temperature Sensor (TS-ADC)
  - Support User-Defined Mode and Automatic Mode
  - In User-Defined Mode, start_of_conversion can be controlled completely by

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software, and also can be generated by hardware.
- In Automatic Mode, the temperature of alarm (high/low temperature) interrupt can be configurable
- In Automatic Mode, the temperature of system reset can be configurable
- -40~125°C temperature range and 1°C temperature resolution

- Successive approximation ADC (SARADC)
  - 10-bit resolution
  - Up to 1MS/s sampling rate
  - 2 single-ended input channels (for RV1106G2, RV1106G3)
  - 3 single-ended input channels (for RV1106)

- OTP
  - Support 8K bits Size, 7K bits for secure application
  - Support Program/Read/Idle mode

- Package Type
  - RV1106G2/G3: RoHS QFN128 (body: 12.3mm x 12.3mm pitch 0.35mm)
  - RV1106: RoHS BGA313 (body: 14mm x 11mm pitch 0.6mm)

1.3 Block Diagram

The following figure shows the basic block diagram.

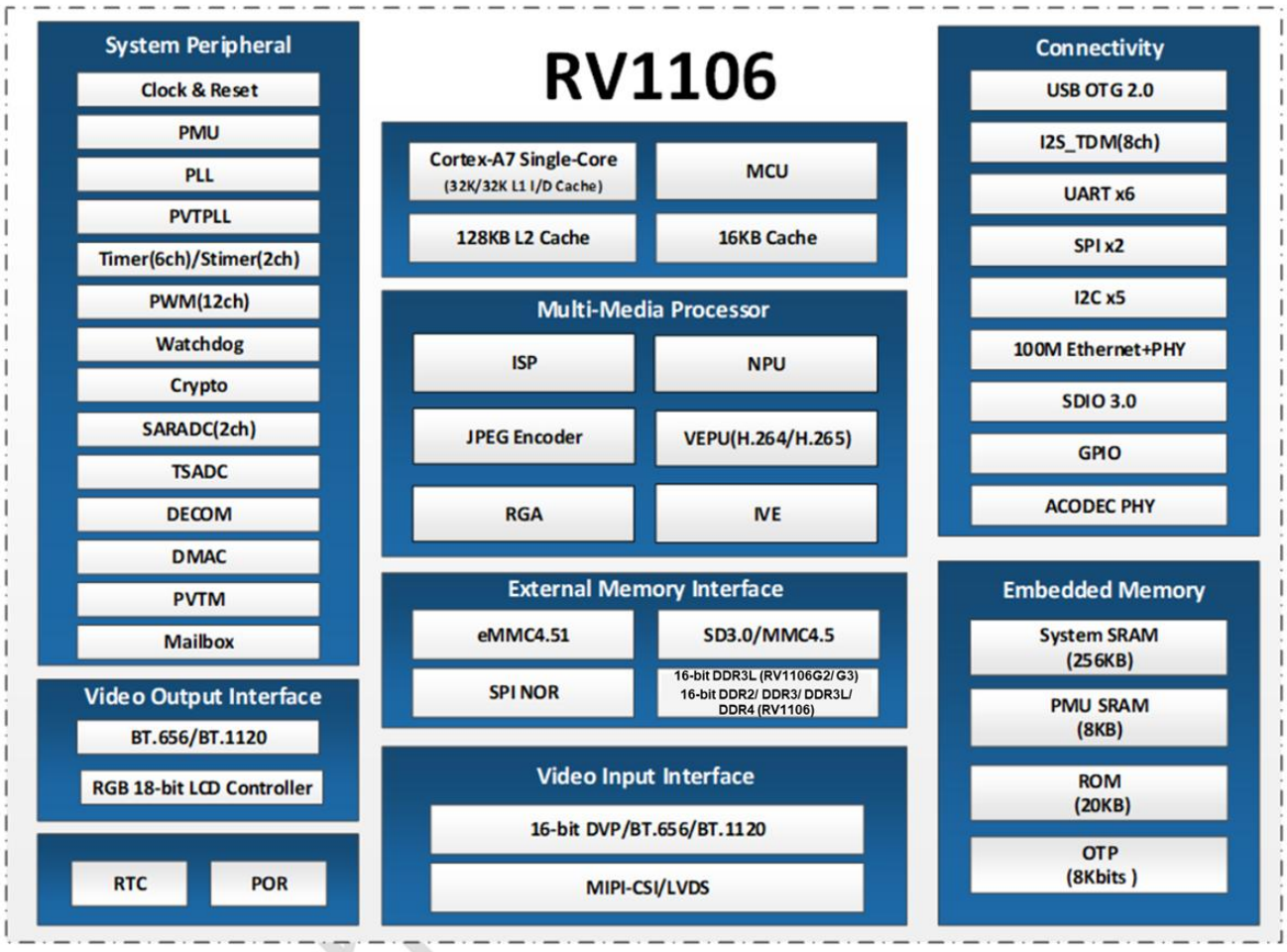


Fig.1-1 Block Diagram

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Chapter 2 Package Information

2.1 Order Information

Orderable Device	RoHS status	Package	Package Q'ty	Device Feature
RV1106G2	RoHS	QFN128	152 pcs per tray 1,520 pcs per inner box 9,120 pcs per outer box	Cortex A7 + MCU + 1.0 TOPS NPU + 1Gb DDR3L + 5 megapixel@30FPS
RV1106G3	RoHS	QFN128	152 pcs per tray 1,520 pcs per inner box 9,120 pcs per outer box	Cortex A7 + MCU + 1.0 TOPS NPU + 2Gb DDR3L + 8 megapixel@15FPS
RV1106	RoHS	BGA313	147 pcs per tray 1,470 pcs per inner box 8.820 pcs per outer box	Cortex A7 + MCU + 1.0 TOPS NPU + 8 megapixel@15FPS

2.2 Top Marking

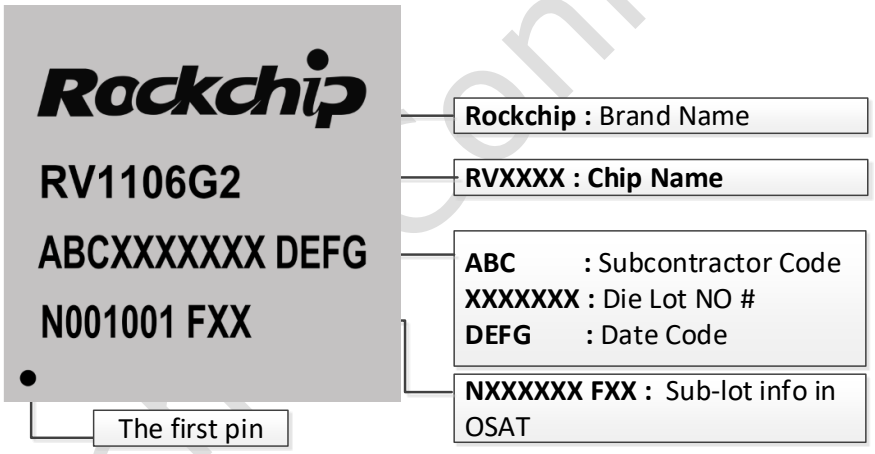
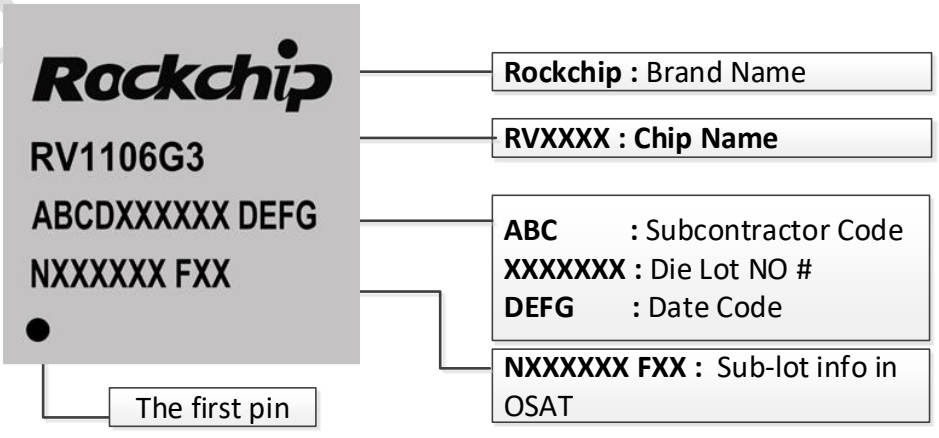


Fig.2-1 RV1106G2 Package Definition



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Fig.2-2 RV1106G3 Package Definition

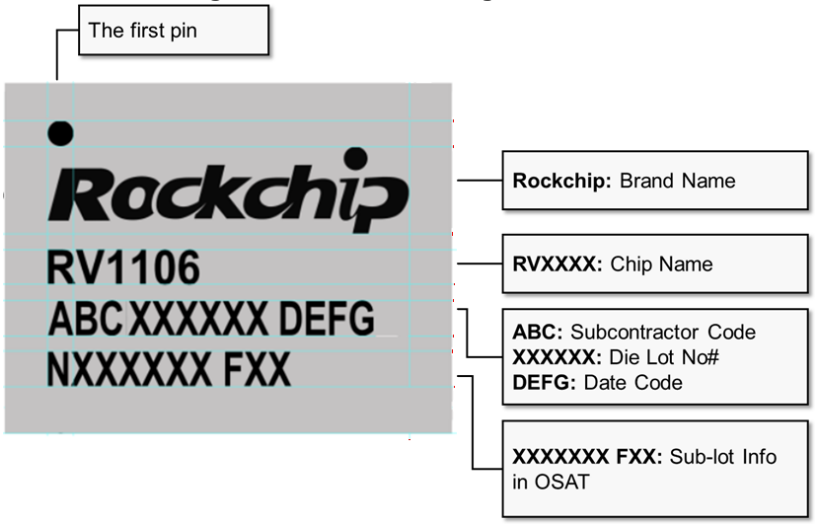


Fig.2-3 RV1106 Package Definition

2.3 Package Dimension

RV1106G2 & RV1106G3

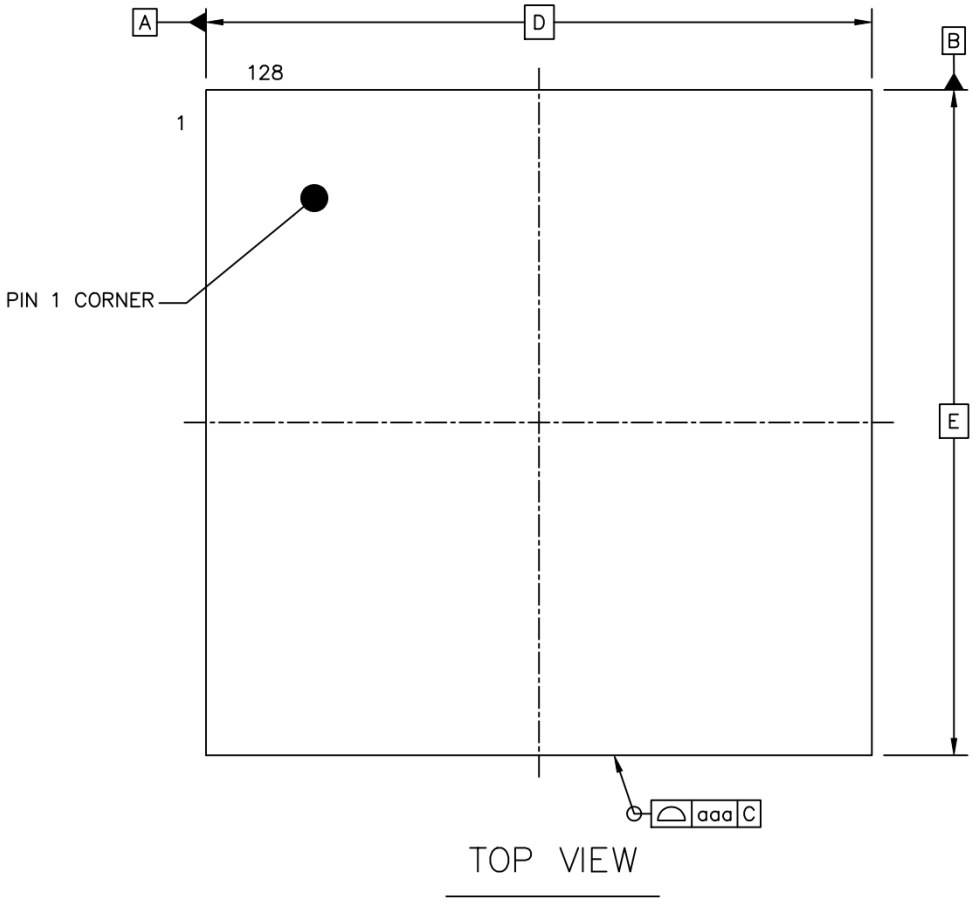


Fig.2-4 Package Top View (RV1106G2 & RV1106G3)

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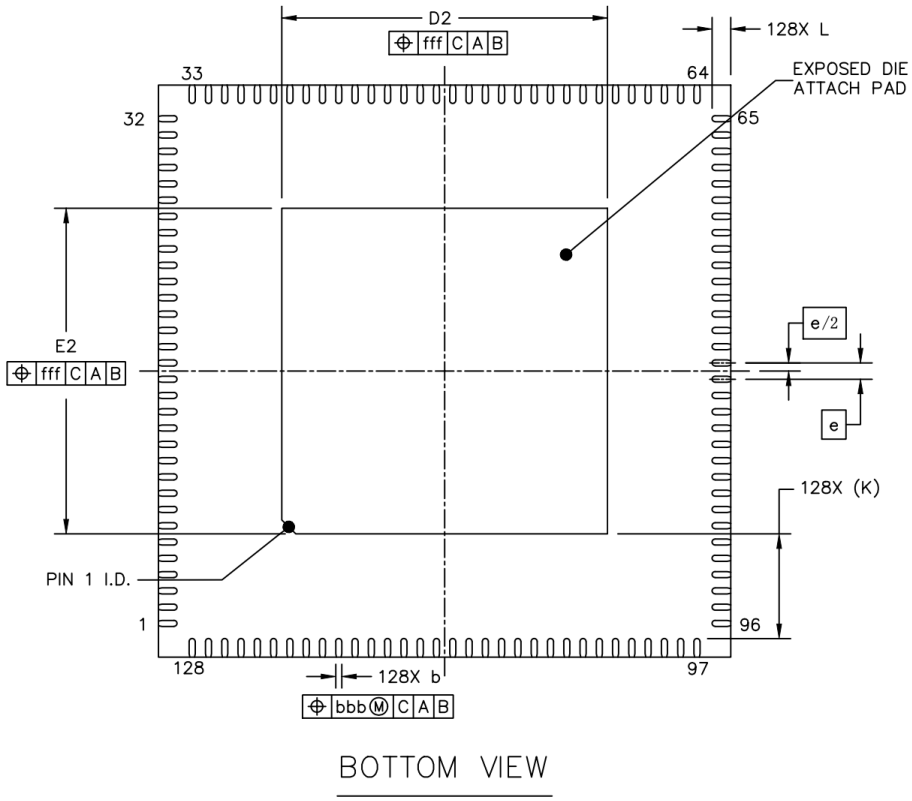


Fig.2-5 Package Bottom View (RV1106G2 & RV1106G3)

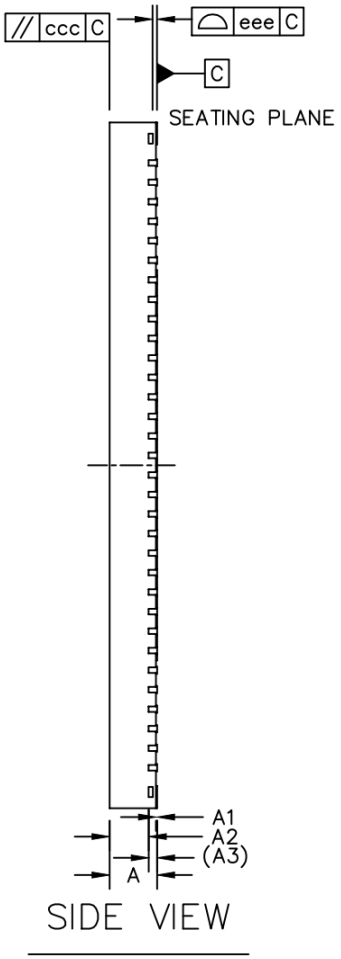


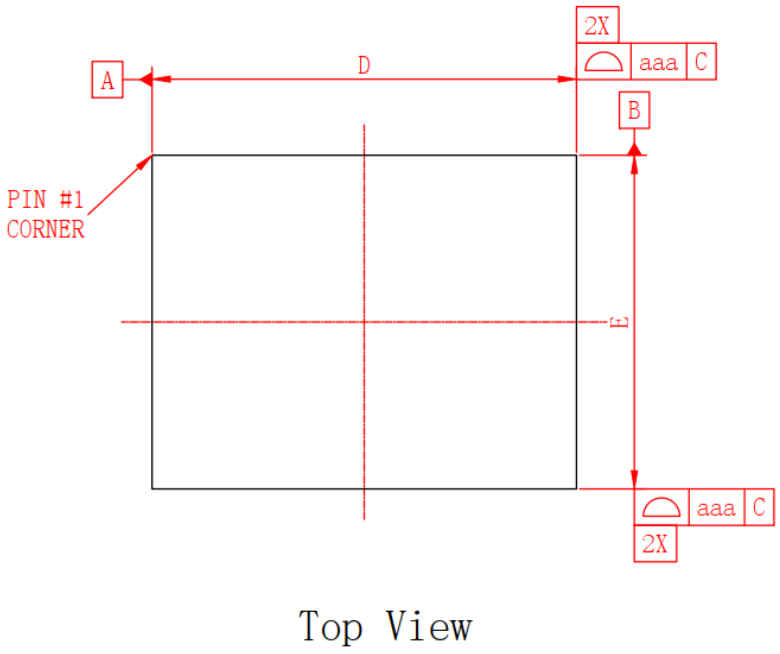
Fig.2-6 Package Side View (RV1106G2 & RV1106G3)

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		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.7	---
L/F THICKNESS		A3		0.152 REF	
LEAD WIDTH		b	0.08	0.13	0.18
BODY SIZE	X	D		12.3 BSC	
	Y	E		12.3 BSC	
LEAD PITCH		e		0.35 BSC	
EP SIZE	X	D2	6.9	7	7.1
	Y	E2	6.9	7	7.1
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K		2.25 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.07	
EXPOSED PAD OFFSET		fff		0.1	

Fig.2-7 Package Dimension (RV1106G2 & RV1106G3)

RV1106



Top View

Fig.2-8 Package Top View (RV1106)

Roc

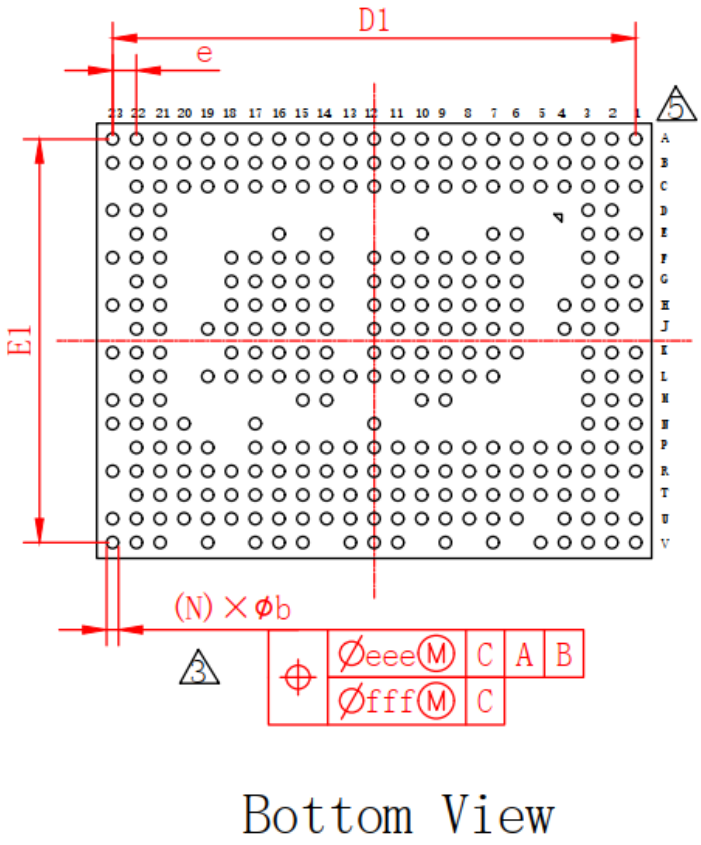


Fig.2-9 Package Bottom View (RV1106)

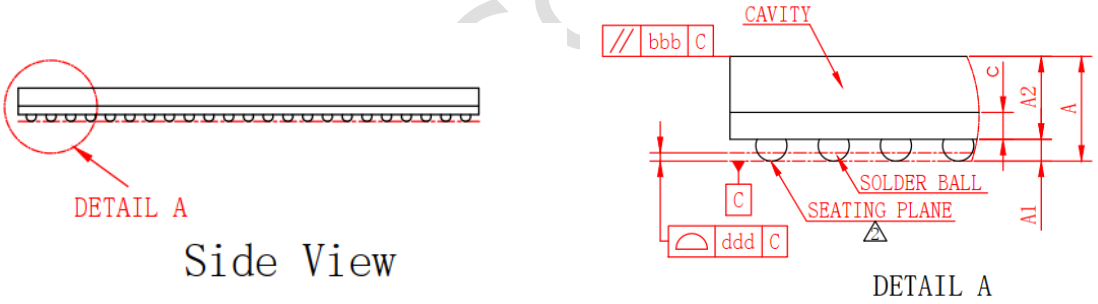


Fig.2-10 Package Side View (RV1106)

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symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	---	---	1.110	---	---	0.044
A1	0.160	0.210	0.260	0.006	0.008	0.010
A2	0.750	0.800	0.850	0.030	0.031	0.033
c	0.230	0.260	0.290	0.009	0.010	0.011
D	13.900	14.000	14.100	0.547	0.551	0.555
E	10.900	11.000	11.100	0.429	0.433	0.437
D1	---	13.200	---	---	0.520	---
E1	---	10.200	---	---	0.402	---
e	---	0.600	---	---	0.024	---
b	0.250	0.300	0.350	0.010	0.012	0.014
aaa	0.100			0.004		
bbb	0.100			0.004		
ddd	0.080			0.003		
eee	0.150			0.006		
fff	0.080			0.003		
Ball Diam	0.300			0.012		
N	313			313		
MD/ME	23/18			23/18		

Fig.2-11 Package Dimension (RV1106)

2.4 MSL Information

Moisture sensitivity level：MSL3

2.5 Lead Finish/Ball Material Information

Lead finish/Ball material：Sn

2.6 Pin Number List

RV1106G2 & RV1106G3

Table 2-1 Pin Number Order Information for RK1106G2 & G3

Pin Name	Pin	Pin Name	Pin
MIPI_AVDD1V8/GPIO7_VCC1V8	1	I2C1_SDA_M0/UART1_CTS_M0/PWM6_M0/GPIO0_A6_d	65
VI_CIF_CLKO_M0/MIPI_CLK0_OUT/GPIO3_C4_d	2	nPOR	66
VI_CIF_VSYNC_M0/GPIO3_C5_d	3	PMU_DVDD0V9	67
VI_CIF_D10/PWM7_IR_M2/MIPI_CLK1_OUT/GPIO3_C6_d	4	OSC_XIN	68
VI_CIF_D11/UART5_TX_M2/I2C4_SCL_M2/GPIO3_C7_d	5	OSC_XOUT	69
VI_CIF_D13/UART5_RTS_M2/I2C3_SCL_M2/GPIO3_D1_d	6	OSC_AVDD1V8/PLL_AVDD1V8	70
VI_CIF_D12/UART5_RX_M2/I2C4_SDA_M2/GPIO3_D0_d	7	OSC_PLL_DVDD	71
VI_CIF_D14/UART5_CTS_M2/I2C3_SDA_M2/GPIO3_D2_d	8	UART3_TX_M0/I2C2_SCL_M0/PWM7_IR_M0/GPIO1_A0_d	72
VI_CIF_D15/PWM1_M2/GPIO3_D3_d	9	UART3_RX_M0/I2C2_SDA_M0/PWM4_M0/GPIO1_A1_d	73
DVDD_1	10	PWM0_M0/CPU_AVS/VI_CIF_D0_M1/GPIO1_A2_d	74
SDMMC0_DET/GPIO3_A1_u	11	UART1_TX_M0/I2C0_SCL_M0/GPIO1_A3_d	75
SDMMC0_D1/UART2_TX_M0/PWM9_M0/GPIO3_A2_u	12	UART1_RX_M0/I2C0_SDA_M0/GPIO1_A4_d	76
GPIO4_VCC	13	UART4_RX_M0/PWM3_IR_M1/GPIO1_B0_d	77
SDMMC0_D0/UART2_RX_M0/PWM8_M0/GPIO3_A3_u	14	UART4_TX_M0/PWM7_IR_M1/SPI1_CS1_M0/VI_CIF_D1_M1/GPIO1_B1_d	78
SDMMC0_CLK/UART5_RTS_M0/I2C0_SCL_M2/JTAG_LPMCU_TCK_M1/PWM10_M0/GPIO3_A4_d	15	JTAG_CPU_TCK_M1/UART2_TX_M1/JTAG_HPMCU_TCK_M0/JTAG_LPMCU_TCK_M0/GPIO1_B2_d	79
SDMMC0_CMD/UART5_CTS_M0/I2C0_SDA_M2/JTAG_LPMCU_TMS_M1/PWM11_IR_M0/GPIO3_A5_u	16	JTAG_CPU_TMS_M1/UART2_RX_M1/JTAG_HPMCU_TMS_M0/JTAG_LPMCU_TMS_M0/GPIO1_B3_u	80
SDMMC0_D3/UART5_TX_M0/JTAG_CPU_TMS_M0/JTAG_HPMCU_TMS_M1/GPIO3_A6_u	17	GPIO1_VCC3V3	81
SDMMC0_D2/UART5_RX_M0/JTAG_CPU_TCK_M0/JTAG_HPMCU_TCK_M1/GPIO3_A7_u	18	DVDD_5	82
RTC_AVDD3V3	19	VO_LCDC_D1/VI_CIF_D8_M1/PWM10_M1/UART4_RTS_M1/GPIO1_C6_d	83
RTC_XOUT	20	VO_LCDC_D0/VI_CIF_D9_M1/PWM11_IR_M1/UART4_CTS_M1/GPIO1_C7_d	84

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Pin Name	Pin	Pin Name	Pin
RTC_XIN	21	VO_LCDC_CLK/VI_CIF_CLKO_M1/I2C3_SCL_M1/UART5_TX_M1/PWM11_IR_M2/AUD_DSM_N/GPIO1_D3_d	85
SARADC_IN1/PWM1_M1/GPI4_C1_z	22	VO_LCDC_VSYNC/VI_CIF_VSYNC_M1/I2C3_SDA_M1/UART5_RX_M1/SPI0_CS1_M0/PWM0_M1/AUD_DSM_P/GPIO1_D2_d	86
SARADC_IN0/GPI4_C0_z	23	VO_LCDC_HSYNC/VI_CIF_HREF_M1/PWM10_M2/UART5_CTS_M1/UART3_RX_M1/GPIO1_D1_d	87
SARADC_USB_AVDD1V8	24	GPIO6_VCC	88
USB_VBUSDET	25	VO_LCDC_DEN/VI_CIF_CLKI_M1/PWM3_IR_M2/UART5_RTS_M1/UART3_TX_M1/GPIO1_D0_d	89
USB_DM	26	VO_LCDC_D2/VI_CIF_D7_M1/PWM9_M1/UART4_TX_M1/SDMMC1_D2_M1/GPIO1_C5_d	90
USB_DP	27	VO_LCDC_D3/VI_CIF_D6_M1/PWM8_M1/UART4_RX_M1/SDMMC1_D3_M1/GPIO1_C4_d	91
USB_AVDD3V3	28	VO_LCDC_D4/VI_CIF_D5_M1/PWM6_M2/I2C4_SDA_M1/SDMMC1_CMD_M1/SPI0_MISO_M0/GPIO1_C3_d	92
CODEC_LINEOUT	29	VO_LCDC_D5/VI_CIF_D4_M1/PWM5_M2/I2C4_SCL_M1/SDMMC1_CLK_M1/SPI0_MOSI_M0/GPIO1_C2_d	93
CODEC_VCM	30	VO_LCDC_D6/VI_CIF_D3_M1/PWM4_M2/SPI0_CLK_M0/SDMMC1_D0_M1/GPIO1_C1_d	94
CODEC_AVDD1V8	31	VO_LCDC_D7/VI_CIF_D2_M1/PWM2_M2/SPI0_CS0_M0/SDMMC1_D1_M1/GPIO1_C0_d	95
CODEC_MICBIAS	32	OTP_AVDD1V8/ETH_AVDD1V8/TSADC_AVDD1V8	96
CODEC_MIC0N	33	ETH_PHY_RXN	97
CODEC_MIC0P	34	ETH_PHY_RXP	98
CODEC_MIC1N	35	ETH_PHY_TXN	99
CODEC_MIC1P	36	ETH_PHY_TXP	100
CODEC_AVSS	37	ETH_AVDD3V3	101
EMMC_D5/SPI1_CLK_M0/UART1_RX_M2/I2C2_SCL_M1/GPIO4_A7_u	38	ETH_EXTR	102
EMMC_D3/FSPI_D3/GPIO4_A6_u	39	DVDD_6	103
EMMC_D4/SPI1_CS0_M0/UART1_TX_M2/I2C2_SDA_M1/GPIO4_A5_u	40	UART0_TX_M1/I2C1_SDA_M1/VO_LCDC_D17/PWM6_M1/GPIO2_B1_d	104
EMMC_D0/FSPI_D0/GPIO4_A4_u	41	UART0_RX_M1/I2C1_SCL_M1/VO_LCDC_D16/PWM5_M1/GPIO2_B0	105
EMMC_D1/FSPI_D1/GPIO4_A3_u	42	UART0_CTS_M1/I2S0_SDO1_SDI3/VO_LCDC_D15/PWM4_M1/I2C3_SDA_M0/PRELIGHT_TRIG_OUT/GPIO2_A7_d	106
GPIO3_VCC	43	UART0_RTS_M1/I2S0_SDO2_SDI2/VO_LCDC_D14/PWM2_M1/I2C3_SCL_M0/FLASH_TRIG_OUT/GPIO2_A6_d	107
EMMC_D2/FSPI_D2/GPIO4_A2_u	44	GPIO5_VCC	108
EMMC_D6/SPI1_MOSI_M0/UART0_TX_M2/I2C0_SCL_M1/GPIO4_A1_u	45	SDMMC1_D1_M0/I2S0_SCLK/VO_LCDC_D8/UART1_CTS_M1/I2C4_SDA_M0/GPIO2_A0_d	109
EMMC_D7/SPI1_MISO_M0/UART0_RX_M2/I2C0_SDA_M1/GPIO4_A0_u	46	SDMMC_D0/I2S0_LRCK/VO_LCDC_D9/UART1_RTS_M1/I2C4_SCL_M0/GPIO2_A1_d	110
EMMC_CMD/FSPI_CS0/GPIO4_B0_u	47	SDMMC1_CLK_M0/I2S0_MCLK/VO_LCDC_D10/GPIO2_A2_d	111
EMMC_CLK/FSPI_CLK/GPIO4_B1_d	48	SDMMC1_CMD_M0/I2S0_SDO3_SDI1/VO_LCDC_D11/GPIO2_A3_d	112
DDR_VDDQ_1	49	SDMMC1_D3_M0/I2S0_SDO0/VO_LCDC_D12/UART1_TX_M1/GPIO2_A4_d	113
DDR_VDDQ_2	50	SDMMC1_D2_M0/I2S0_SDI0/VO_LCDC_D13/UART1_RX_M1/GPIO2_A5_d	114
DVDD_2	51	CPU_DVDD	115
DRAM_ZQ	52	DVDD_7	116
DDR_PLL_AVDD1V8	53	VI_CIF_D0_M0/MIPI_CSI_RX_D3N/LVDS_RX_D3N/GPI3_B0_d	117
DVDD_3	54	VI_CIF_D1_M0/MIPI_CSI_RX_D3P/LVDS_RX_D3P/GPI3_B1_d	118
DVDD_4	55	VI_CIF_D2_M0/MIPI_CSI_RX_CK1N/LVDS_RX_CK1N/GPI3_B2_d	119
DDR_VDDQ_3	56	VI_CIF_D3_M0/MIPI_CSI_RX_CK1P/LVDS_RX_CK1P/GPI3_B3_d	120
TVSS	57	VI_CIF_D4_M0/MIPI_CSI_RX_D2N/LVDS_RX_D2N/GPI3_B4_d	121
UART0_RX_M0/CLK_32K/CLK_REFOUT/RTC_CLKO/GPIO0_A0_z	58	VI_CIF_D5_M0/MIPI_CSI_RX_D2P/LVDS_RX_D2P/GPI3_B5_d	122
UART0_TX_M0/PWM2_M0/GPIO0_A1_d	59	VI_CIF_D6_M0/MIPI_CSI_RX_D1N/LVDS_RX_D1N/GPI3_B6_d	123
PWM3_IR_M0/GPIO0_A2_d	60	VI_CIF_D7_M0/MIPI_CSI_RX_D1P/LVDS_RX_D1P/GPI3_B7_d	124
PMU_VCC3V3	61	VI_CIF_D8_M0/MIPI_CSI_RX_CK0N/LVDS_RX_CK0N/GPI3_C0_d	125
PMIC_PWR_CTRL_M1/GPIO0_A3_u	62	VI_CIF_D9_M0/MIPI_CSI_RX_CK0P/LVDS_RX_CK0P/GPI3_C1_d	126
PMIC_PWR_CTRL_M0/PWM1_M0/GPIO0_A4_d	63	VI_CIF_CLKI_M0/MIPI_CSI_RX_D0N/LVDS_RX_D0N/GPI3_C2_d	127
I2C1_SCL_M0/UART1_RTS_M0/PWM5_M0/GPIO0_A5_d	64	VI_CIF_HREF_M0/MIPI_CSI_RX_D0P/LVDS_RX_D0P/GPI3_C3_d	128
		VSS	E-PAD

RV1106

Table 2-2 Pin Number Order Information for RK1106

Pin Name	Pin	Pin Name	Pin
OSC_SOC_XOUT	K23	DDR_DQ0	U14
OSC_SOC_XIN	K22	DDR_DQ1	U13

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Pin Name	Pin	Pin Name	Pin
PLL_DVDD0V9	J19	DDR_DQ2	V12
OSC_PLL_AVDD1V8	J21	DDR_DQ3	U12
OTP_TSADC_AVDD1V8	C18	DDR_DQ4	P10
OSC_RTC_XIN	H1	DDR_DQ5	U11
OSC_RTC_XOUT	H2	DDR_DQ6	P9
RTC_AVDD3V3	G2	DDR_DQ7	V11
nPOR_DET	L22	DDQ_DM0	R9
UART0_RX_M0/CLK_32K_IN/REF_CLK_OUT/RTC_CLK_OUT/GPIO0_A0_z	P22	DDR_DQS0P	T10
UART0_TX_M0/PWM2_M0/GPIO0_A1_d	P21	DDR_DQS0N	U10
PWM3_IR_M0/GPIO0_A2_d	N22	DDR_DQ8	U8
PWR_CTRL_M1/GPIO0_A3_u	N23	DDR_DQ9	T7
PWR_CTRL_M0/PWM1_M0/GPIO0_A4_d	N21	DDR_DQ10	V7
I2C1_SCL_M0/UART1_RTS_M0/PWM5_M0/GPIO0_A5_d	M23	DDR_DQ11	U7
I2C1_SDA_M0/UART1_CTS_M0/PWM6_M0/GPIO0_A6_d	M22	DDR_DQ12	R7
PMU_DVDD0V9/OSC_AVDD0V9	L19	DDR_DQ13	T5
PMU_VCC3V3	M21	DDR_DQ14	V5
UART3_TX_M0/I2C2_SCL_M0/PWM7_IR_M0/GPIO1_A0_d	J22	DDR_DQ15	T4
UART3_RX_M0/I2C2_SDA_M0/PWM4_M0/GPIO1_A1_d	H23	DDQ_DM1	R5
PWM0_M0/CPU_AVS/VI_CIF_D0_M1/GPIO1_A2_d	H22	DDR_DQS1P	T6
UART1_TX_M0/I2C0_SCL_M0/GPIO1_A3_d	H21	DDR_DQS1N	U6
UART1_RX_M0/I2C0_SDA_M0/GPIO1_A4_d	G22	DDR2_A3/DDR3_A0/DDR4_A0	U21
UART4_RX_M0/PWM3_IR_M1/GPIO1_B0_d	G21	DDR2_A4/DDR3_A1/DDR4_A1	U20
UART4_TX_M0/PWM7_IR_M1/SPI1_CS1_M0/VI_CIF_D1_M1/GPIO1_B1_d	F23	DDR2_A7/DDR3_A2/DDR4_A2	U22
JTAG_CPU_TCK_M1/UART2_TX_M1/JTAG_HPMCU_TCK_M0/JTAG_LPMCU_TCK_M0/GPIO1_B2_d	F22	DDR2_A1/DDR3_A3/DDR4_A3	V21
JTAG_CPU_TMS_M1/UART2_RX_M1/JTAG_HPMCU_TMS_M0/JTAG_LPMCU_TMS_M0/GPIO1_B3_u	E22	DDR2_A2/DDR3_A4/DDR4_A4	V19
VCCIO1_VCC3V3	F21	DDR2_A5/DDR3_A5/DDR4_A5	V22
SARADC_IN0/GPIO4_C0_z	J2	DDR2_A6/DDR3_A6/DDR4_A6	U19
SARADC_IN1/PWM1_M1/GPIO4_C1_z	J3	DDR2_A9/DDR3_A7/DDR4_A7	T22
VCCIO2_VCC1V8	H4	DDR2_A8/DDR3_A8/DDR4_A8	T19
EMMC_D0/FSPI_D0/GPIO4_A4_u	T3	DDR2_A12/DDR3_A9/DDR4_A9	R23
EMMC_D1/FSPI_D1/GPIO4_A3_u	U1	DDR2_RAS/DDR3_A10/DDR4_A10	U16
EMMC_D2/FSPI_D2/GPIO4_A2_u	U2	DDR2_A11/DDR3_A11/DDR4_A11	R20
EMMC_D3/FSPI_D3/GPIO4_A6_u	R3	DDR2_A0/DDR3_A12/DDR4_A12	T18
EMMC_D4/SPI1_CS0_M0/UART1_TX_M2/I2C2_SDA_M1/GPIO4_A5_u	T2	DDR2_A13/DDR3_A13/DDR4_A13	U23
EMMC_D5/SPI1_CLK_M0/UART1_RX_M2/I2C2_SCL_M1/GPIO4_A7_u	R2	DDR2_A14/DDR3_A14/DDR4_A14	R13
EMMC_D6/SPI1_MOSI_M0/UART0_TX_M2/I2C0_SCL_M1/GPIO4_A1_u	V2	DDR2_A15/DDR3_A15/DDR4_A15	U17
EMMC_D7/SPI1_MISO_M0/UART0_RX_M2/I2C0_SDA_M1/GPIO4_A0_u	U3	DDR2_BA1/DDR3_WE/DDR4_A16	R15
EMMC_CMD/FSPI_CS0/GPIO4_B0_u	V3	DDR2_BA0/DDR3_CAS/DDR4_A17	V17
EMMC_CLK/FSPI_CLK/GPIO4_B1_d	U4	DDR2_WE/DDR3_RAS/DDR4_ACT	R14
VCCIO3_VCC	P5	DDR2_A10/DDR3_BA0/DDR4_BA0	R17
SDMMC_DET/GPIO3_A1_u	D3	DDR2_CAS/DDR3_BA1/DDR4_BA1	U18
SDMMC_D0/UART2_RX_M0/PWM8_M0/GPIO3_A3_u	E3	DDR2_BA2/DDR3_BA2/DDR4_BG0	R18
SDMMC_D1/UART2_TX_M0/PWM9_M0/GPIO3_A2_u	D2	DDR4_BG1	P16
SDMMC_D2/UART5_RX_M0/JTAG_CPU_TCK_M0/JTAG_HPMCU_TCK_M1/GPIO3_A7_u	F2	DDR2_CLKP/DDR3_CLKP/DDR4_CLKP	U15
SDMMC_D3/UART5_TX_M0/JTAG_CPU_TMS_M0/JTAG_HPMCU_TMS_M1/GPIO3_A6_u	F3	DDR2_CLKN/DDR3_CLKN/DDR4_CLKN	V15
SDMMC_CLK/UART5_RTS_M0/I2C0_SCL_M2/JTAG_LPMCU_TCK_M1/PWM10_M0/GPIO3_A4_d	E1	DDR2_CKE0/DDR3_CKE0/DDR4_CKE0	T12
SDMMC_CMD/UART5_CTS_M0/I2C0_SDA_M2/JTAG_LPMCU_TMS_M1/PWM11_IR_M0/GPIO3_A5_u	E2	DDR2_CKE1/DDR3_CKE1/DDR4_CKE1	P13
VCCIO4_VCC	G3	DDR2_ODT0/DDR3_ODT0/DDR4_ODT0	V16
SDIO_D0_M0/I2S0_LRCK/VO_LCDC_D9/UART1_RTS_M1/I2C4_SCL_M0/GPIO2_A1_d	A13	DDR2_ODT1/DDR3_ODT1/DDR4_ODT1	P17
SDIO_D1_M0/I2S0_SCLK/VO_LCDC_D8/UART1_CTS_M1/I2C4_SDA_M0/GPIO2_A0_d	B14	DDR2_CS0/DDR3_CS0/DDR4_CS0	T15
SDIO_D2_M0/I2S0_SDI0/VO_LCDC_D13/UART1_RX_M1/GPIO2_A5_d	A11	DDR2_CS1/DDR3_CS1/DDR4_CS1	P14
SDIO_D3_M0/I2S0_SDO0/VO_LCDC_D12/UART1_TX_M1/GPIO2_A4_d	B12	DDR3_RSTN/DDR4_RSTN	R22
SDIO_CLK_M0/I2S0_MCLK/VO_LCDC_D10/GPIO2_A2_d	B13	DDR_ZQ	L12
SDIO_CMD_M0/I2S0_SDO3_SDI1/VO_LCDC_D11/GPIO2_A3_d	A12	DDR_PLL_AVDD1V8	N12
UART0_RTS_M1/I2S0_SDO2_SDI2/VO_LCDC_D14/PWM2_M1/I2C3_SCL_M0/FLASH_TRIGOUT/GPIO2_A6_d	A14	DDR_PLL_AVSS	R12
UART0_CTS_M1/I2S0_SDO1_SDI3/VO_LCDC_D15/PWM4_M1/I2C3_SDA_M0/PRELIGHT_TRIGOUT/GPIO2_A7_d	B15	DDR_VREF_OUT	N17
UART0_RX_M1/I2C1_SCL_M1/VO_LCDC_D16/PWM5_M1/GPIO2_B0_d	A15	DDR_VDDQ_1	M9
UART0_TX_M1/I2C1_SDA_M1/VO_LCDC_D17/PWM6_M1/GPIO2_B1_d	B16	DDR_VDDQ_2	M10
VCCIO5_VCC	C13	DDR_VDDQ_3	M14
VI_BT656_D0_M1/VO_LCDC_D7/VI_CIF_D2_M1/PWM2_M2/SPI0_CS0_M0/SDIO_D1_M1/GPIO1_C0_d	A20	DDR_VDDQ_4	M15
VI_BT656_D1_M1/VO_LCDC_D6/VI_CIF_D3_M1/PWM4_M2/SPI0_CLK_M0/SDIO_D0_M1/GPIO1_C1_d	B20	VSS_1	A1
VI_BT656_D2_M1/VO_LCDC_D5/VI_CIF_D4_M1/PWM5_M2/I2C4_SCL_M1/SDIO_CLK_M1/SPI0_MOSI_M0/GPIO1_C2_d	A21	VSS_2	A4
VI_BT656_D3_M1/VO_LCDC_D4/VI_CIF_D5_M1/PWM6_M2/I2C4_SDA_M1/SDIO_CMD_M1/SPI0_MISO_M0/GPIO1_C3_d	A22	VSS_3	A7

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Pin Name	Pin	Pin Name	Pin
VI_BT656_D4_M1/VO_LCDC_D3/VI_CIF_D6_M1/PWM8_M1/UART4_RX_M1/SDIO_D3_M1/GPIO1_C4_d	C20	VSS_4	A10
VI_BT656_D5_M1/VO_LCDC_D2/VI_CIF_D7_M1/PWM9_M1/UART4_TX_M1/SDIO_D2_M1/GPIO1_C5_d	B21	VSS_5	A16
VI_BT656_D6_M1/VO_LCDC_D1/VI_CIF_D8_M1/PWM10_M1/UART4_RTS_M1/GPIO1_C6_d	D23	VSS_6	A19
VI_BT656_D7_M1/VO_LCDC_D0/VI_CIF_D9_M1/PWM11_IR_M1/UART4_CTS_M1/GPIO1_C7_d	D21	VSS_7	A23
VI_BT656_CLK_M1/VO_LCDC_DEN/VI_CIF_CLKI_M1/PWM3_IR_M2/UART5_RTS_M1/UART3_TX_M1/GPIO1_D0_d	B22	VSS_8	B11
VO_LCDC_HSYNC/VI_CIF_HREF_M1/PWM10_M2/UART5_CTS_M1/UART3_RX_M1/GPIO1_D1_d	B23	VSS_9	C1
VO_LCDC_VSYNC/VI_CIF_VSYNC_M1/I2C3_SDA_M1/UART5_RX_M1/SPI0_CS1_M0/PWM0_M1/AUD_DSM_P/GPIO1_D2_d	C22	VSS_10	C6
VO_LCDC_CLK/VI_CIF_CLKO_M1/I2C3_SCL_M1/UART5_TX_M1/PWM11_IR_M2/AUD_DSM_N/GPIO1_D3_d	D22	VSS_11	C8
VCCIO6_VCC	B19	VSS_12	C9
VI_BT1120_D0/VI_CIF_D0_M0/MIPI_CSI_RX_D3N/LVDS_RX_D3N/GPI3_B0_d	C10	VSS_13	C11
VI_BT1120_D1/VI_CIF_D1_M0/MIPI_CSI_RX_D3P/LVDS_RX_D3P/GPI3_B1_d	B10	VSS_14	C17
VI_BT1120_D2/VI_CIF_D2_M0/MIPI_CSI_RX_CLK1N/LVDS_RX_CLK1N/GPI3_B2_d	A9	VSS_15	C19
VI_BT1120_D3/VI_CIF_D3_M0/MIPI_CSI_RX_CLK1P/LVDS_RX_CLK1P/GPI3_B3_d	B9	VSS_16	C21
VI_BT1120_D4/VI_CIF_D4_M0/MIPI_CSI_RX_D2N/LVDS_RX_D2N/GPI3_B4_d	C7	VSS_17	E21
VI_BT1120_D5/VI_CIF_D5_M0/MIPI_CSI_RX_D2P/LVDS_RX_D2P/GPI3_B5_d	B7	VSS_18	F6
VI_BT1120_D6/VI_CIF_D6_M0/MIPI_CSI_RX_D1N/LVDS_RX_D1N/GPI3_B6_d	B8	VSS_19	F7
VI_BT1120_D7/VI_CIF_D7_M0/MIPI_CSI_RX_D1P/LVDS_RX_D1P/GPI3_B7_d	A8	VSS_20	F8
VI_BT656_D0_M0/VI_BT1120_D8/VI_CIF_D8_M0/MIPI_CSI_RX_CLK0N/LVDS_RX_CLK0N/GPI3_C0_d	A6	VSS_21	F9
VI_BT656_D1_M0/VI_BT1120_D9/VI_CIF_D9_M0/MIPI_CSI_RX_CLK0P/LVDS_RX_CLK0P/GPI3_C1_d	B6	VSS_22	F10
VI_BT656_CLK_M0/VI_BT1120_CLK/VI_CIF_CLKI_M0/MIPI_CSI_RX_D0N/LVDS_RX_D0N/GPI3_C2_d	B5	VSS_23	F11
VI_CIF_HREF_M0/MIPI_CSI_RX_D0P/LVDS_RX_D0P/GPI3_C3_d	A5	VSS_24	F12
VI_CIF_CLKO_M0/MIPI_CLK0_OUT/GPIO3_C4_d	B4	VSS_25	F14
VI_CIF_VSYNC_M0/GPIO3_C5_d	A3	VSS_26	F15
VI_BT656_D2_M0/VI_BT1120_D10/VI_CIF_D10/PWM7_IR_M2/MIPI_CLK1_OUT/GPIO3_C6_d	B3	VSS_27	F16
VI_BT656_D3_M0/VI_BT1120_D11/VI_CIF_D11/UART5_TX_M2/I2C4_SCL_M2/GPIO3_C7_d	A2	VSS_28	G1
VI_BT656_D4_M0/VI_BT1120_D12/VI_CIF_D12/UART5_RX_M2/I2C4_SDA_M2/GPIO3_D0_d	B2	VSS_29	G7
VI_BT656_D5_M0/VI_BT1120_D13/VI_CIF_D13/UART5_RTS_M2/I2C3_SCL_M2/GPIO3_D1_d	C2	VSS_30	G8
VI_BT656_D6_M0/VI_BT1120_D14/VI_CIF_D14/UART5_CTS_M2/I2C3_SDA_M2/GPIO3_D2_d	C3	VSS_31	G9
VI_BT656_D7_M0/VI_BT1120_D15/VI_CIF_D15/PWM1_M2/GPIO3_D3_d	B1	VSS_32	G10
VCCIO7_VCC1V8	C4	VSS_33	G11
MIPI_AVDD1V8	C5	VSS_34	G12
FEPHY_RXP	B18	VSS_35	G14
FEPHY_RXN	A18	VSS_36	G15
FEPHY_TXP	A17	VSS_37	G16
FEPHY_TXN	B17	VSS_38	G17
FEPHY_REXT	C15	VSS_39	H3
FEPHY_AVDD1V8	C14	VSS_40	H6
FEPHY_AVDD3V3	C16	VSS_41	H7
CODEC_MIC0P	M2	VSS_42	H8
CODEC_MIC0N	N1	VSS_43	H9
CODEC_MIC1P	P2	VSS_44	H10
CODEC_MIC1N	P1	VSS_45	H11
CODEC_LINEOUT	M1	VSS_46	H12
CODEC_VCM	M3	VSS_47	H14
CODEC_MICBIAS	N2	VSS_48	H15
CODEC_AVDD1V8	N3	VSS_49	H16
CODEC_AVSS_1	L3	VSS_50	H17
CODEC_AVSS_2	R1	VSS_51	H18
USB_DP	L1	VSS_52	J7
USB_DM	L2	VSS_53	J8
USB_ID	K6	VSS_54	J9
USB_VBUSDET	K2	VSS_55	J10
USB_AVDD1V8	J4	VSS_56	J11
USB_AVDD3V3	K3	VSS_57	J12
CPU_DVDD	C12	VSS_58	J14
DVDD_1	E6	VSS_59	J15
DVDD_2	E7	VSS_60	J16
DVDD_3	E10	VSS_61	J17
DVDD_4	E14	VSS_62	J18
DVDD_5	E16	VSS_63	K7
DVDD_6	G18	VSS_64	K8
DVDD_7	L8	VSS_65	K9

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Pin Name	Pin	Pin Name	Pin
DVDD_8	L13	VSS_66	K10
DVDD_9	L14	VSS_67	K11
		VSS_68	K12
		VSS_69	K14
		VSS_70	K15
		VSS_71	K16
		VSS_72	K17
		VSS_73	K18
		VSS_74	K21
		VSS_75	L7
		VSS_76	L9
		VSS_77	L10
		VSS_78	L11
		VSS_79	L15
		VSS_80	L16
		VSS_81	L17
		VSS_82	L18
		VSS_83	L21
		VSS_84	P3
		VSS_85	P4
		VSS_86	P6
		VSS_87	P7
		VSS_88	P8
		VSS_89	P11
		VSS_90	P12
		VSS_91	P15
		VSS_92	P19
		VSS_93	P20
		VSS_94	R4
		VSS_95	R6
		VSS_96	R8
		VSS_97	R10
		VSS_98	R11
		VSS_99	R16
		VSS_100	R19
		VSS_101	R21
		VSS_102	T8
		VSS_103	T9
		VSS_104	T11
		VSS_105	T13
		VSS_106	T14
		VSS_107	T16
		VSS_108	T17
		VSS_109	T20
		VSS_110	T21
		VSS_111	U9
		VSS_112	V1
		VSS_113	V4
		VSS_114	V9
		VSS_115	V13
		VSS_116	V23
		AVSS	K1
		TVSS	N20
		NC1	G6
		NC2	F17
		NC3	F18
		NC4	J6

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Chapter 3 Electrical Specification

3.1 Absolute Ratings

The below table provides the absolute ratings.

Absolute maximum or minimum ratings specify the values beyond which the device may be damaged permanently. Long-term exposure to absolute maximum ratings conditions may affect device reliability.

Table 3-1 Absolute Ratings

Parameters	Related Power Group	Min	Max	Unit
Supply voltage for CPU	CPU_DVDD	0	1.15	V
Supply voltage for LOGIC	DVDD_*i*(*i*=1~7)	0	0.99	V
Supply voltage for PMU	PMU_DVDD0V9	0	0.99	V
Supply voltage for DDR IO	DDR_VDDQ_i(i=1~4)	-0.3	1.94	V
Supply voltage for RTC	RTC_AVDD3V3	0	3.6	V
0.9V supply voltage	DVDD_*i*(*i*=1~7) PMU_DVDD0V9 OSC_PLL_DVDD	0	0.99	V
1.8V/3.3V supply voltage	GPIO*i*_VCC(*i*=3~6, 1.8V/3.3V mode)	0	3.63	V
1.8V supply voltage	OSC_AVDD1V8/PLL_AVDD1V8 MIPI_AVDD1V8/GPIO7_VCC1V8 SARADC_USB_AVDD1V8 DDR_PLL_AVDD1V8 OTP_AVDD1V8/ETH_AVDD1V8/TSADC_AVDD1V8 CODEC_AVDD1V8	0	1.98	V
3.3V supply voltage	GPIO1_VCC3V3 PMU_VCC3V3 RTC_AVDD3V3 USB_AVDD3V3 ETH_AVDD3V3	0	3.63	V
Storage Temperature	Tstg	-40	125	℃
Max Conjunction Temperature	Tj	NA	125	℃

3.2 Recommended Operating Conditions

The following table describes the recommended operating conditions.

Table 3-2 Recommended Operating Conditions

Parameters	Symbol	Min	Typ	Max	Unit
Voltage for CPU	CPU_DVDD	0.85	0.90	1.05	V
Voltage for LOGIC	DVDD_*i*(*i*=1~7)	0.81	0.90	0.95	V
Voltage for PMU	PMU_DVDD0V9	0.81	0.90	0.95	V
DDR2 IO VDDQ Power	DDR_VDDQ	1.7	1.8	1.9	V
DDR3 IO VDDQ Power	DDR_VDDQ	1.425	1.5	1.575	V
DDR3L IO VDDQ Power	DDR_VDDQ	1.283	1.35	1.45	V
DDR4 IO VDDQ Power	DDR_VDDQ	1.14	1.2	1.26	V
Voltage for RTC	RTC_AVDD3V3	1.6	3.3	3.6	V
Voltage for PLL Analog (1.8V)	OSC_AVDD1V8/PLL_AVDD1V8	1.62	1.8	TBD	V
Voltage for GPIO (1.8V/3.3V)	GPIO*i*_VCC(*i*=3~6)	1.62 3.0	1.8 3.3	1.98 3.465	V
Voltage for GPIO (1.8V only)	MIPI_AVDD1V8/GPIO7_VCC1V8	1.62	1.8	1.98	V
Voltage for GPIO (3.3V only)	GPIO1_VCC3V3 PMU_VCC3V3	3.0	3.3	3.465	V
Voltage for USB/SARADC Analog (1.8V)	SARADC_USB_AVDD1V8	1.62	1.8	1.98	V
Voltage for USB Analog (3.3V)	USB_AVDD3V3	3.0	3.3	3.6	V
Voltage for OTP/MAC Analog (1.8V)	OTP_AVDD1V8/ETH_AVDD1V8/TSADC_AVDD1V8	1.62	1.8	1.98	V
Voltage for MAC Analog (3.3V)	ETH_AVDD3V3	2.97	3.3	3.63	V
Voltage for CODEC Analog	CODEC_AVDD1V8	1.62	1.8	1.98	V

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Parameters	Symbol	Min	Typ	Max	Unit
Voltage for MIPI Analog	MIPI_AVDD1V8/GPIO7_VCC1V8	1.62	1.8	1.98	V
Voltage for DDR PHY PLL	DDR_PLL_AVDD1V8	TBD	1.8	TBD	V
Ambient Operating Temperature	T_A	-20	25	85	℃

3.3 DC Characteristics

Table 3-3 DC Characteristics

Parameters		Symbol	Min	Typ	Max	Unit
Digital GPIO @3.3V	Input Low Voltage	Vil	-0.3	NA	0.8	V
	Input High Voltage	Vih	2.0	NA	VDDO+0.3	V
	Output Low Voltage	Vol	-0.3	NA	0.4	V
	Output High Voltage	Voh	2.4	NA	VDDO+0.3	V
	Pullup Resistor	Rpu	16	25	43	Kohm
	Pulldown Resistor	Rpd	16	25	43	Kohm
Digital GPIO @1.8V	Input Low Voltage	Vil	-0.3	NA	0.35*VDDO	V
	Input High Voltage	Vih	0.65*VDDO	NA	VDDO+0.3	V
	Output Low Voltage	Vol	-0.3	NA	0.4	V
	Output High Voltage	Voh	1.4	NA	VDDO+0.3	V
	Pullup Resistor	Rpu	16	25	43	Kohm
	Pulldown Resistor	Rpd	16	25	43	Kohm

Parameters		Symbol	Min	Typ	Max	Unit
MIPI_LVDS Combo IO@LVDS HS receiver mode	Common-mod voltage HS receive mode	VCMRX(DC)	0.8	NA	1.32	V
	Differential input high threshold	VIDTH	NA	NA	70	mV
	Differential input low threshold	VIDTL	-70	NA	NA	mV
	Single-ended input high voltage	VIHHS	NA	NA	1.5	V
	Single-ended input low voltage	VILHS	-40	NA	NA	mV
	Differential input impedance	ZID	80	100	125	ohm
MIPI_LVDS Combo IO@ MIPI HS receiver mode	Common-mod voltage HS receive mode	VCMRX(DC)	70	NA	300	mV
	Differential input high threshold	VIDTH	NA	NA	70	mV
	Differential input low threshold	VIDTL	-70	NA	NA	mV
	Single-ended input high voltage	VIHHS	NA	NA	460	mV
	Single-ended input low voltage	VILHS	-40	NA	NA	mV
	Single-ended threshold for HS termination enable	VTERM-EN	NA	NA	450	mV
	Differential input impedance	ZID	80	100	125	ohm
MIPI_LVDS Combo IO@ MIPI LP receiver mode	Logic 1 input voltage	VIH	880	NA	NA	mV
	Logic 0 input voltage, not in ULP State	VIL	NA	NA	550	mV
	Logic 0 input voltage, ULP State	VIL-ULPS	NA	NA	300	mV
	Input hysteresis	VHYST	25	NA	NA	mV
MIPI_LVDS Combo IO@ 1.8V TTL RX mode	Logic 1 input voltage	VIH	1.2	NA	1.58	V
	Logic 0 input voltage, not in ULP State	VIL	NA	NA	0.6	V
	Input hysteresis	VHYST	25	NA	NA	mV

3.4 Electrical Characteristics for General IO

Table 3-4 Electrical Characteristics for Digital General IO

Parameters		Symbol	Test condition	Min	Typ	Max	Unit
Digital GPIO @3.3V	Input leakage current	Ii	Vin = 3.3V or 0V	NA	NA	10	uA
	Tri-state output leakage current	Ioz	Vout = 3.3V or 0V	NA	NA	10	uA

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Parameters		Symbol	Test condition	Min	Typ	Max	Unit
	High level input current	Iih	Vin = 3.3V, pulldown disabled	NA	NA	10	uA
			Vin = 3.3V, pulldown enabled	NA	NA	10	uA
	Low level input current	Iil	Vin = 0V, pullup disabled	NA	NA	10	uA
			Vin = 0V, pullup enabled	NA	NA	10	uA
Digital GPIO @1.8V	Input leakage current	Ii	Vin = 1.8V or 0V	NA	NA	10	uA
	Tri-state output leakage current	Ioz	Vout = 1.8V or 0V	NA	NA	10	uA
	High level input current	Iih	Vin = 1.8V, pulldown disabled	NA	NA	10	uA
			Vin = 1.8V, pulldown enabled	NA	NA	10	uA
	Low level input current	Iil	Vin = 0V, pullup disabled	NA	NA	10	uA
			Vin = 0V, pullup enabled	NA	NA	10	uA

Note: VDDO and DVDD are both IO power Supply

3.5 Electrical Characteristics for PLL

Table 3-5 Electrical Characteristics for INT PLL

Parameters		Symbol	Test condition	Min	Typ	Max	Unit
	Input clock frequency(Frac)	F_{in}	Fin = FREF @1.8V/0.99V	10	NA	800	MHz
	VCO operating range	F_{vco}	Fvco = Fref * FBDIV @3.3V/0.99V	475	NA	1900	MHz
	Output clock frequency	F_{out}	Fout = Fvco/POSTDIV @3.3V/0.99V	9	NA	1900	MHz
	Lock time	T_{lt}	@ 3.3V/0.99V, FREF=24M,REFDIV=1	NA	1000	1500	Input clock cycles

Table 3-6 Electrical Characteristics for FRAC PLL

Parameters		Symbol	Test condition	Min	Typ	Max	Unit
	Input clock frequency(Frac)	F_{in}	Fin = FREF @1.8V/0.99V	1	NA	1200	MHz
	VCO operating range	F_{vco}	Fvco = Fref * FBDIV @3.3V/0.99V	950	NA	3800	MHz
	Output clock frequency	F_{out}	Fout = Fvco/POSTDIV @3.3V/0.99V	19	NA	3800	MHz
	Lock time	T_{lt}	@ 3.3V/0.99V, FREF=24M,REFDIV=1	NA	250	500	Input clock cycles

Notes:

① *REFDIV is the input divider value;*
② *FBDIV is the feedback divider value;*
③ *POSTDIV is the output divider value*

3.6 Electrical Characteristics for USB2.0 Interface

Table 3-7 Electrical Characteristics for USB2.0 Interface

Parameters	Symbol	Test condition	Min	Typ	Max	Unit
Transmitter						
Output resistance	ROUT	Classic mode (Vout = 0 or 3.3V)	40.5	45	49.5	ohms
		HS mode (Vout = 0 to 800mV)	40.5	45	49.5	ohms
Output Capacitance	COUT	seen from D+ or D-			3	pF
Output Common Mode Voltage	VM	Classic (LS/FS) mode	1.45	1.65	1.85	V
		HS mode	0.175	0.2	0.225	V
Differential output signal high	VOH	Classic (LS/FS); Io=0mA	2.97	3.3	3.63	V

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Parameters	Symbol	Test condition	Min	Typ	Max	Unit
		Classic (LS/FS); Io=6mA	2.2	2.7	NA	V
		HS mode; Io=0mA	360	400	440	mV
Differential output signal low	VOL	Classic (LS/FS); Io=0mA	-0.33	0	0.33	V
		Classic (LS/FS); Io=6mA	NA	0.3	0.8	V
		HS mode; Io=0mA	-40	0	40	mV
Receiver						
Receiver sensitivity	RSENS	Classic mode	NA	+-250	NA	mV
		HS mode	NA	+-25	NA	mV
Receiver common mode	RCM	Classic mode	0.8	1.65	2.5	V
		HS mode (differential and squelch comparator)	0.1	0.2	0.3	V
		HS mode (disconnect comparator)	0.5	0.6	0.7	V
Input capacitance (seen at D+ or D-)			NA	NA	3	pF
Squelch threshold			100	NA	150	mV
Disconnect threshold			570	600	664	mV

3.7 Electrical Characteristics for MIPI CSI interface

Table 3-8 HS Receiver AC specifications (for MIPI mode)

Parameters	Symbol	Min	Typ	Max	Unit
Common-mode interference beyond 450 MHz	ΔVCMRX(HF)	NA	NA	100	mV
Common-mode interference 50MHz – 450MHz	ΔVCMRX(LF)	-50	NA	50	mV
Common-mode termination	CCM	NA	NA	60	pF

Table 3-9 LP Receiver AC specifications (for MIPI mode)

Parameters	Symbol	Min	Typ	Max	Unit
Input pulse rejection	eSPIKE	NA	NA	300	V.ps
Minimum pulse width response	TMIN-RX	20	NA	NA	ns
Peak interference amplitude	VINT	NA	NA	200	mv
Interference frequency	fINT	450	NA	NA	MHz

Table 3-10 HS Receiver AC Specifications (for LVDS mode)

Parameters	Symbol	Min	Typ	Max	Unit
Common-mode interference beyond 450 MHz	ΔVCMRX(HF)	NA	NA	100	mV
Common-mode interference 50MHz – 450MHz	ΔVCMRX(LF)	-50	NA	50	mV
Common-mode termination	CCM	NA	NA	50	pF

3.8 Electrical Characteristics for Audio CODEC interface

Table 3-11 Electrical Characteristics for Audio CODEC

Test conditions: AVDD = 1.8V, DVDD = 0.8V, TA = 25°C, 1KHz Sine Input, Fs = 48KHz

Parameters	Symbol	Test condition	Min	Typ	Max	Units
Operating Condition						
Analog Supply	AVDD		1.62	1.8	1.98	V
Microphone Bias						
Bias Voltage	V_{MICB}		0.8*AVDD	NA	0.975*AVDD	V
Bias Current	I_{MICB}		NA	NA	3	mA
Microphone Gain Boost PGA						
Programmable Gain	G_{BST}		0	NA	20	dB
Gain Step Size			NA	20	NA	dB
Input Resistance	R_{IN}	G_{BST}=0dB	NA	44	NA	KΩ

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Parameters	Symbol	Test condition	Min	Typ	Max	Units
		G_{BST}=20dB	NA	8	NA	KΩ
Input Capacitance	C_{IN}		NA	10	NA	pF
ALC PGA						
Programmable Gain	G_{ALC}		-9	NA	37.5	dB
Gain Step Size			NA	1.5	NA	dB
ADC						
Signal to Noise Ratio	SNR	A-weighted	NA	92	NA	dB
Total Harmonic Distortion	THD	-3dBFS input	NA	-80	NA	dB
Channel Separation			NA	80	NA	dB
Power Supply Rejection	PSRR	1KHz	NA	80	NA	dB
Digital Filter Pass Band Ripple			0.1	0.125	0.125	
DAC Line Output						
Programmable Gain	G_{DRV}		-39	NA	6	dB
Gain Step Size			NA	1.5	NA	dB
Signal to Noise Ratio	SNR	A-weighted	NA	93	NA	dB
Total Harmonic Distortion	THD	-3dBFS output 600Ω load	NA	-84	NA	dB
Power Supply Rejection	PSRR	1KHz	NA	55	NA	dB
Power Consumption						
Standby			NA	0.01	NA	mA
Mono Recording			NA	2.5	NA	mA
Mono Playback		Quiescent output	NA	2.5	NA	mA

3.9 Electrical Characteristics for SARADC

Table 3-12 Electrical Characteristics for SARADC

Parameters	Symbol	Test condition	Min	Typ	Max	Unit
Resolution			NA	10	NA	bit
Effective Number of Bit	ENOB		NA	9	NA	bit
Differential Non-Linearity	DNL		-1	NA	+1	LSB
Integral Non-Linearity	INL		-2	NA	+2	LSB
Reference voltage	VREFP		NA	1.8	NA	V
Input Capacitance	C_{IN}		NA	8	NA	pF
Sampling Rate	f_S		NA	NA	1	MS/s
Spurious Free Dynamic Range	SFDR	f_S=1MS/s f_{OUT}=1.17KHz	NA	61	NA	dB
Signal to Noise and Harmonic Ratio	SNDR		NA	56	NA	dB

3.10 Electrical Characteristics for TSADC

Table 3-13 Electrical Characteristics for TSADC

Parameters	Symbol	Test condition	Min	Typ	Max	Unit
Accuracy from -40℃ to 125℃	T_{JACC}		NA	NA	±3	℃
Sensing Temperature Range	T_{RANGE}		-40	NA	125	℃
Resolution	T_{LSB}		NA	0.6	NA	℃

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Chapter 4 Thermal Management

4.1 Overview

For reliability and operability concerns, the absolute maximum junction temperature has to be below 125℃.

4.2 Package Thermal Characteristics

Table 4-1 provides the thermal resistance characteristics for the package used on the SoC. The resulting simulation data for reference only, please prevail in kind test.

Table 4-1 Thermal Resistance Characteristics

Parameter	Symbol	Typical	Unit
Junction-to-ambient thermal resistance	$\boldsymbol{\theta_{JA}}$	27.3① 25.2②	(℃/$\boldsymbol{W}$)
Junction-to-board thermal resistance	$\boldsymbol{\theta_{JB}}$	12.5① 10.81②	(℃/$\boldsymbol{W}$)
Junction-to-case thermal resistance	$\boldsymbol{\theta_{JC}}$	8.4① 4.8②	(℃/$\boldsymbol{W}$)

*Note: The JEDEC 2S2P PCB is 4 layers, 114.3mm*76.2mm.*

1. *Data refers to thermal resistance of RV1106G2 and RV1106G3*
2. *Data refers to thermal resistance of RV1106*

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